

Project 1: CVDs predictions (Team ML212)

Leila Benjelloun

leila.benjelloun@epfl.ch

Rania Hatrouhou

rania.hatrouhou@epfl.ch

Ismael Berrada

ismael.berrada@epfl.ch

1 Introduction

This project addresses the challenge of predicting cardiovascular diseases (CVDs) from large-scale health survey data. The dataset comprises 328,135 samples with 321 features from telephone health surveys, presenting a severely imbalanced binary classification problem with only 8.83% positive CVD cases. This imbalance, combined with extensive missing data and heterogeneous feature types, requires careful methodological choices to achieve meaningful predictive performance.

2 Methodology

2.1 Data Understanding and Preprocessing Strategy

Our exploratory analysis revealed a dataset with 91.17% negative cases, extensive missingness (239 features affected, up to 99.99% missing), and mixed data types: 101 categorical survey responses, 73 continuous measurements, and 70 binary features. We identified survey-specific missing codes (7, 9, 77, 99) and removed 8 non-predictive date/ID features.

Rather than applying a single preprocessing approach, we designed a systematic pipeline with increasing sophistication levels to understand the trade-offs between data retention and feature quality. This approach was motivated by the hypothesis that different preprocessing strategies would significantly impact model performance on this challenging dataset.

2.2 Experimental Design: A Systematic Three-Phase Approach

Given the strong class imbalance in our dataset, we use the F1-score (see [A.2](#)), as our primary evaluation metric, as it provides a balanced measure of precision and recall and more accurately reflects model performance on the minority class.

Phase 1: Preprocessing Impact Assessment We fixed logistic regression as our baseline model and systematically evaluated preprocessing levels 0-5 using 5-fold stratified cross-validation [check detailed levels [1](#)].

Phase 2: Model Selection with Fixed Preprocessing

After identifying the best preprocessing level, we compared logistic regression, regularized logistic regression, and random forest using comprehensive grid search. This phase tested our hypothesis that different model architectures would perform differently on the preprocessed data.

Phase 3: Alternative Imbalance Handling Recognizing that our levels use oversampling, we explored cost-sensitive learning with Level 1 (no oversampling) preprocessing to compare different approaches to class imbalance. This phase addressed the question of whether algorithmic solutions (class weights) could match data-level solutions (oversampling).

Phase 4: Fine-Tuning In the final phase, we focused on fine-tuning the best-performing model identified during our comparison stage. This step involved refining key hyper-parameters to maximize predictive performance while preventing overfitting.

3 Results and Analysis

3.1 Preprocessing Impact: From Failure to Success

As shown in [Figure 1](#), performance improved markedly with more structured preprocessing, with Level 5 achieving the best overall training results across all metrics (F1 = 0.676, ROC-AUC = 0.858, PR-AUC = 0.744, Precision = 0.711, Recall = 0.644). Compared to Level 2, which relied on target encoding, the shift to one-hot encoding in Level 5 yielded substantial gains in both discriminative power and stability. This suggests that a simpler, non-parametric encoding of categorical variables may generalize better under this data distribution.

Consequently, we adopt Level 5 as our standardized preprocessing configuration for the subsequent analyses.

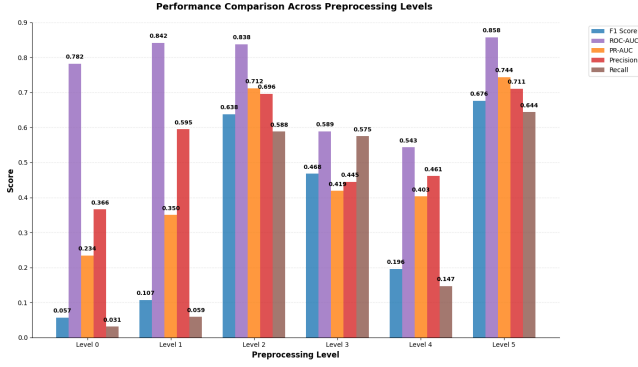


Figure 1: Performance comparison across preprocessing levels (L0–L5) showing F1-Score, ROC-AUC, PR-AUC, Precision, and Recall metrics.

3.2 Model Architecture Comparison

With Level 5 preprocessing stabilized, model comparison revealed clear performance differences: regularized and standard logistic regression performed best ($F1 = 0.423$ – 0.424), far surpassing random forest ($F1 = 0.365$). The L2 regularization ($\lambda = 0.001$) offered the best bias–variance trade-off, limiting overfitting without sacrificing accuracy.

Despite random forest’s higher training precision (0.723), its weak recall led to poorer F1-scores, indicating that ensemble methods were less suited to this data-preprocessing setup.

Given its strong generalization potential, we selected the regularized logistic model, as further fine-tuning could yield even better performance.

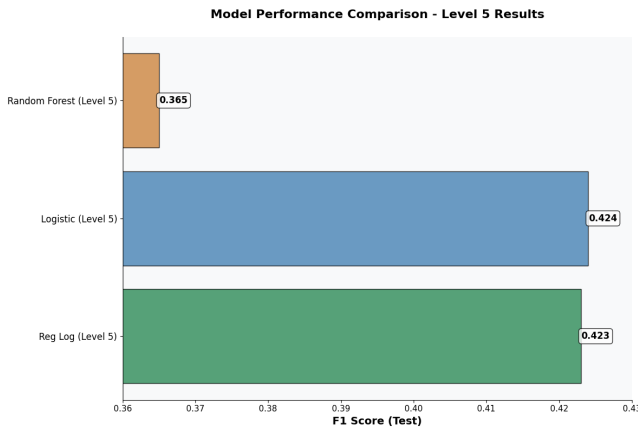


Figure 2: Performance comparison across test sets of different models showing obtained F1 scores on Alcrowd.

3.3 Class Imbalance Handling: Oversampling vs. Cost-Sensitive Learning

Figure 3 compares two strategies for addressing class imbalance: oversampling (Level 5) and cost-sensitive learning using Level 1. The regularized logistic model trained on Level 5 preprocessing achieved the highest F1

score (0.423), outperforming both weighted logistic and weighted regularized logistic approaches. This suggests that effective preprocessing and balanced feature representation had a stronger impact on handling imbalance than weighting schemes alone.

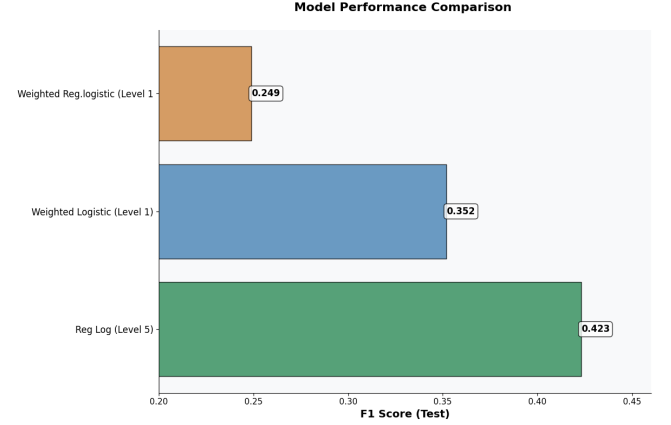


Figure 3: Performance comparison across test sets of different weighted models showing obtained F1 scores on Alcrowd.

3.4 Final Model Optimization

Given that regularized logistic regression with Level 5 preprocessing demonstrated the best performance, we proceeded to conduct an extensive hyperparameter optimization to further refine the model.

Grid Search Configuration: The grid search was performed using 5-fold cross-validation on the training set, exploring the following hyperparameter ranges:

Hyperparameter	Values Tested
Learning rate (γ)	{0.05, 0.08, 0.10, 0.15}
Regularization strength (λ)	{0.0003, 0.0005, 0.001, 0.002, 0.005}
Maximum iterations (<i>max_iters</i>)	3000

This grid search configuration surprisingly yielded the same performance, achieving an **F1-score of 0.423** and an **accuracy of 0.843**, rising questions on the effectiveness of fine-tuning regularized logistic regression.

4 Conclusion

Our results highlight that (1) preprocessing quality is the main driver of model performance, (2) L2 regularization offers the best bias–variance balance, and (3) oversampling outperforms cost-sensitive learning for imbalance. Moreover, one-hot encoding proved more robust than target encoding, confirming that in imbalanced medical data, preprocessing choices often matter more than model complexity.

A Appendix

A.4 Precision–Recall Trade-off Analysis

A.1 Preprocessing Levels Detailed

Table 1: Preprocessing configurations across levels (L0–L5).

Level	Missing $\leq$	Drop low var	Lowvar th.	Cont. impute	Cat. impute	Drop dups	One-hot encodi
L0	0.99	✗	–	mean	mode	✗	✗
L1	0.80	✓	0.999	mean	mode	✗	✗
L2	0.60	✓	0.995	mean	mode	✗	✓
L3	0.40	✓	0.990	median	mode	✓	✓
L4	0.60	✓	0.995	mean	mode	✓	✓
L5	0.50	✗	–	mean	mode	✗	✓

Level	Target Encoding	Aggressiveness level	One-hot encoding
L0	✗	–	✗
L1	✓	light	✗
L2	✓	medium	✗
L3	✓	aggressive	✗
L4	✓	medium–aggressive	✗
L5	✗	–	✓

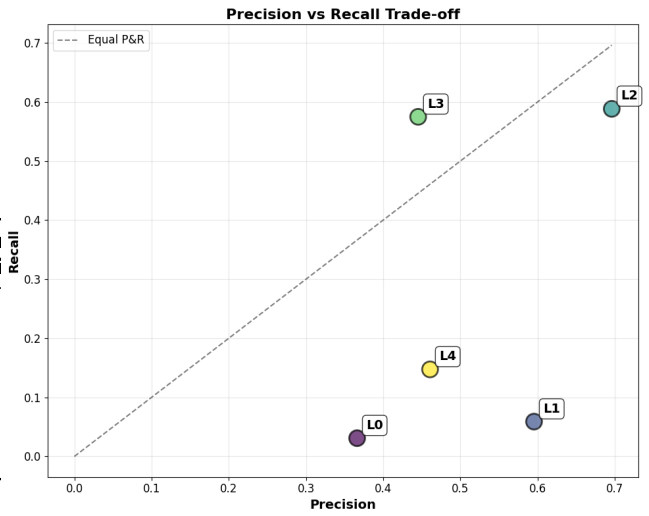


Figure 4: Precision vs. Recall trade-off across preprocessing levels (L0–L4).

A.2 F1-Score

The F1-score is the harmonic mean of precision and recall, defined as:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (1)$$

A.3 Model Performance Comparison Table

Table 2: Model Performance Comparison Across Preprocessing Levels During Training

Model	Preproc.	F1	PR-AUC	ROC-AUC	Prec.
Raw Data Results					
Log. Regression	Raw	0.000	0.089	0.503	0.000
Reg. Logistic	Raw	0.000	0.089	0.503	0.000
Level 5 Preprocessing Results					
Log. Regression	L5	0.676	0.745	0.858	0.711
Reg. Logistic	L5	0.676	0.744	0.857	0.710
Random Forest	L5	0.546	0.691	0.824	0.723
Level 1 + Weighted Results					
Weighted Reg. Log.	L1	0.245	0.358	0.845	0.505
Weighted Log.	L1	0.305	0.412	0.840	0.564